

ActuaryGPT — Underwriting Report

Confidential actuarial risk assessment & premium recommendation report.

1. Customer & Policy Profile

Age (Normalized)	65.0	Gender	Male
Height (cm)	170.0	Weight (kg)	110.0
BMI	38.0	Product Info 2	D3
Occupation	Construction	Annual Income	\$45,000.00
Insurance Type	Life	Coverage Amount	\$200,000.00

2. Risk Analysis & Premium Pricing

Predicted Risk Class	Class 8 of 8
Risk Category	High Risk
Recommended Annual Premium	\$40,000.00

3. AI Actuarial Analysis

Actuarial Report: Life Insurance Application Evaluation

Applicant ID: [N/A - Current Application]

Date: October 26, 2023

Insurance Type: Life

Coverage Amount: \$200,000.00

1. Risk Assessment

The applicant presents a highly elevated risk profile across multiple dimensions. Key risk factors include:

* **Age (65):** Significantly increases mortality risk.

* **BMI (38.0):** Falls into Class III Obesity, indicating severe health risks (e.g., cardiovascular disease, diabetes, certain cancers), which substantially elevates mortality.

* **Smoker (Yes):** Smoking is a major health hazard, dramatically increasing the risk of various diseases and premature death.

* **Occupation (Construction):** This is a physically demanding and inherently high-risk occupation due to potential accidents, injuries, and long-term health issues associated with the work environment.

* **Previous Claims (2):** Two previous claims indicate a history of adverse events, which is a significant red flag and suggests a higher propensity for future claims or existing unmitigated risks.

* **Family History (1):** Indicates a positive family history for significant medical conditions, contributing to elevated genetic predisposition risks.

* **Exercise (0) & Alcohol (2):** Lack of exercise combined with moderate/high alcohol consumption further exacerbates health risks associated with obesity and smoking.

* **Gender (Male):** Generally associated with slightly higher mortality rates compared to females at similar ages.

In summary, the applicant exhibits a convergence of advanced age, severe obesity, active smoking, a high-risk occupation, and a history of claims, culminating in a critically high overall risk of mortality.

2. Comparative Analysis

The current application has been predicted to be in Risk Class 8, with an associated premium of \$40,000.00. This is significantly higher than the top similar historical approved cases:

* **POL-2626 (Risk Class 2, Premium \$12,000.00, Similarity 42.8%):** The current applicant is vastly riskier. A Risk Class 2 typically represents a standard or preferred risk profile with minimal adverse factors. The current applicant's profile (age 65, BMI 38, smoker, construction, 2 claims) clearly deviates from such a low-risk profile, justifying the substantial difference in risk class and premium. The stated similarity is likely based on a subset of features that do not fully capture the severity of the current applicant's overall risk.

* **POL-25755 (Risk Class 5, Premium \$20,000.00, Similarity 39.6%):** While Risk Class 5 indicates moderate risk, it is still considerably lower than the current applicant's Risk Class 8. POL-25755 likely presented some adverse factors (e.g., elevated BMI, mild health condition, or moderate age), but not the severe combination of risks seen in the current application.

* **POL-31852 (Risk Class 7, Premium \$30,000.00, Similarity 38.5%):** This case is the closest in terms of risk class and premium, suggesting POL-31852 also had a very high-risk profile. However, the current applicant's assignment to Risk Class 8 and a higher premium (\$40,000 vs \$30,000) indicates that the current applicant's cumulative risk factors are even more severe or numerous than those present in POL-31852. The combination of extreme obesity, smoking, high-risk occupation, and multiple previous claims likely pushes the current applicant into the maximum risk category.

The similarities noted by the RAG context (in the 38-42% range) are relatively low, suggesting these historical cases are not extremely close matches to the current applicant's complex and high-risk profile. The significant difference in predicted risk class and premium, despite some "similarity," highlights the unique severity of the current application.

3. Anomaly & Fraud Detection

Several factors present as potential anomalies or warrant further scrutiny:

* **Extreme Risk Factor Combination:** The confluence of Class III obesity (BMI 38), active smoking, age 65, and a high-risk occupation (Construction) is highly unusual and indicative of an extremely elevated mortality risk.

* **Previous Claims (2):** Two previous claims for a life insurance applicant is a significant red flag. It implies a history of adverse events, which could be related to health issues, accidents, or other circumstances that directly impact mortality risk. This requires detailed investigation into the nature and timing of these claims.

* **Coverage Amount vs. Income:** A \$200,000 coverage amount for an income of \$45,000 represents a coverage multiple of approximately 4.4x. While not inherently fraudulent, this is on the higher end, especially for an individual with such a high-risk profile, and warrants confirmation of genuine need and financial justification.

* **Low Prediction Confidence (33.0%):** Despite assigning the applicant to the maximum Risk Class 8, the Machine Learning Classifier's confidence level is exceptionally low. This is a critical anomaly. It suggests the model is highly uncertain about this specific prediction, potentially due to the uniqueness or extremity of the applicant's combined features which may not have clear analogues in the training data, or conflicting signals within the data. This low confidence mandates a thorough manual review to validate the model's output and premium calculation.

4. Premium Recommendation

The calculated annual premium of \$40,000.00 is high but appears justifiable given the applicant's predicted Risk Class 8 and the severity of identified risk factors.

* **Maximum Risk Class:** Assignment to Risk Class 8 signifies the highest possible mortality risk, necessitating the highest premium loading.

* **Cumulative Risk:** The combined effect of extreme obesity, active smoking, advanced age, high-risk occupation, family history, sedentary lifestyle, and a history of previous claims creates an exceptionally high probability of early mortality, warranting a substantial premium to cover the actuarial risk.

* **Coverage Amount:** The \$200,000 coverage, while possibly high relative to income, represents a significant potential payout that must be adequately priced given the applicant's risk.

* **Historical Comparison:** The premium is higher than the closest historical case (POL-31852, Risk Class 7, \$30,000), which is consistent with the applicant being an even higher risk than that historical example.

The low prediction confidence (33.0%) raises concerns about the *precision* of the \$40,000 figure. While the *direction* (high premium) is undoubtedly correct given the risk profile, the exact value requires further validation.

5. Underwriting Decision Recommendation

****Recommendation: Refer for Manual Underwriting Review****

A direct approval or rejection is not advisable at this stage. The application presents a complex and extremely high-risk profile, coupled with a critically low machine learning prediction confidence. A comprehensive manual review is essential to:

1. ****Validate Risk Factors:**** Thoroughly verify all stated risk factors (BMI, smoking status, previous claims details, occupation specifics, family history).
2. ****Medical Underwriting:**** Require full medical examinations, blood tests, and potentially attending physician statements (APS) to assess the true extent of health risks associated with obesity, smoking, and age.
3. ****Previous Claims Investigation:**** Detailed investigation into the nature, cause, and outcome of the two previous claims is paramount to understand their impact on current and future mortality risk.
4. ****Occupational Hazard Assessment:**** A detailed review of the specific duties within 'Construction' and their associated risks.
5. ****Address Low ML Confidence:**** The underwriting team should critically evaluate the ML model's output and premium, potentially recalibrating based on expert judgment and more comprehensive data gathered during the manual review. This might lead to an adjusted premium or a decline if the risk is deemed uninsurable.
6. ****Financial Underwriting:**** Confirm the financial justification for the \$200,000 coverage given the income and high-risk profile.

Given the totality of the circumstances, the risk is severe, and the automated assessment needs expert human validation.

4. Sign-off & Recommendation

Based on the predictive model output and AI underwriting guidelines, the premium recommendation is formally calculated at **\$40,000.00** per annum. The underwriting team should perform secondary manual verification if the risk class exceeds Class 5.